

Assignment 1 — Univariate and Multivariate Statistics

FinTech 545 — Quantitative Risk Management

September 2026

1 Reading the Shape of a Sample

1.1 Predict

First four moments of `problem1.csv`, using the bias corrected estimators from Week 1:

Moment	Value
n	1000
Mean	+0.00110439
Variance	0.0000962482
Standard deviation	0.00981062
Skewness	−0.6698
Excess kurtosis	+2.3576

The population versions are -0.6688 and $+2.3398$, so at $n = 1000$ the bias correction changes nothing. The check that excess kurtosis $\geq \text{skewness}^2 - 2$ passes, $+2.3576 \geq -1.5513$, so the pair is admissible.

Two facts matter: the sample is **left skewed** and it has **fat tails**.

Part (a).

Normal: ruled out by the fourth moment. Its excess kurtosis is 0 at every μ and σ and the sample is at $+2.36$. The third moment rules it out too, since the normal is symmetric and the sample skew is -0.67 . Neither is a free parameter, so no choice of μ and σ fixes it.

Lognormal: ruled out by the third moment. Its skewness $(e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$ is strictly positive for every $\sigma > 0$ and cannot be made negative. The support is a second objection: the lognormal lives on $(0, \infty)$ and 406 of the 1000 observations are negative.

Student's t: ruled out by the third moment. The t is symmetric, so its skewness is 0 whenever it exists. It handles the fourth moment fine, $6/(\nu - 4) = 2.36$ at $\nu \approx 6.5$, so it gets the tails right and the asymmetry wrong.

NIG: survives. It is the only one of the four that can carry a negative skew and positive excess kurtosis at the same time. Inverting the four moment formulas confirms the solution is inside the parameter space rather than just plausible in principle. Writing $u = \delta\gamma$ and $\rho = \beta/\alpha$, the table

gives $\hat{\mu}_3 = 3\rho/\sqrt{u}$ and $\hat{\mu}_4 = 3/u + \frac{4}{3}\hat{\mu}_3^2$, which solve to $u = 1.705 > 0$ and then to $\alpha = 145.47$, $\beta = -42.41$, $\delta = 0.012254$, $\mu = 0.004840$, with $|\beta| < \alpha$ as required.

1.2 Fit

Matching the mean and variance gives $N(0.00110439, 0.0000962482)$, whose 1% quantile is -0.02171852 .

Part (b). 26 observations fall below it. 10 should.

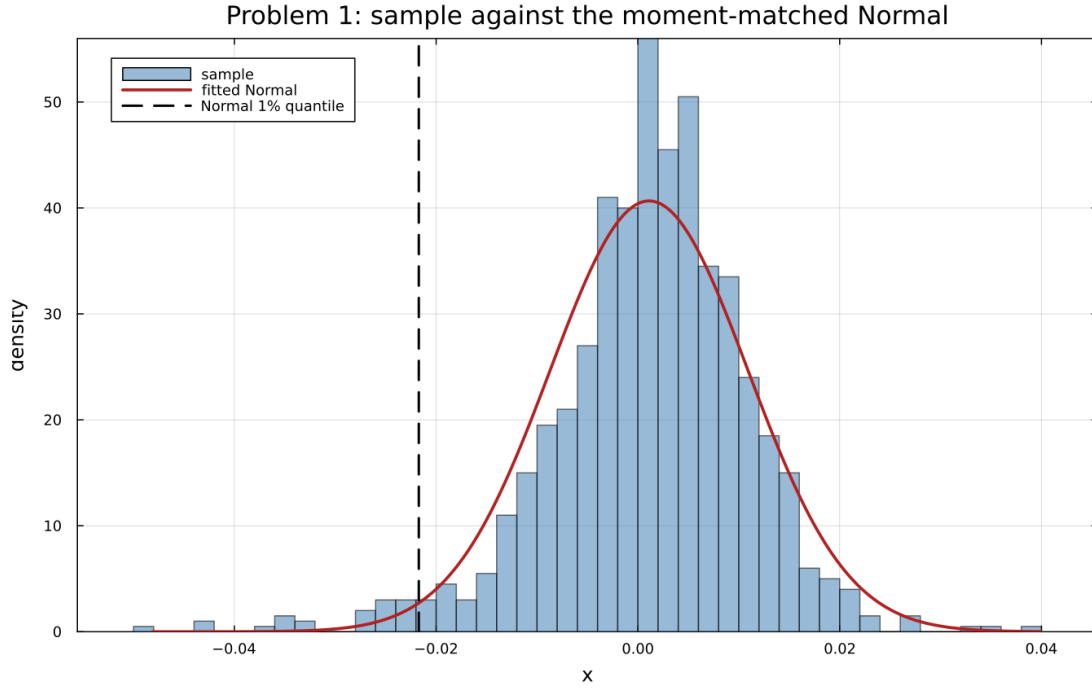


Figure 1: Sample against the moment matched Normal

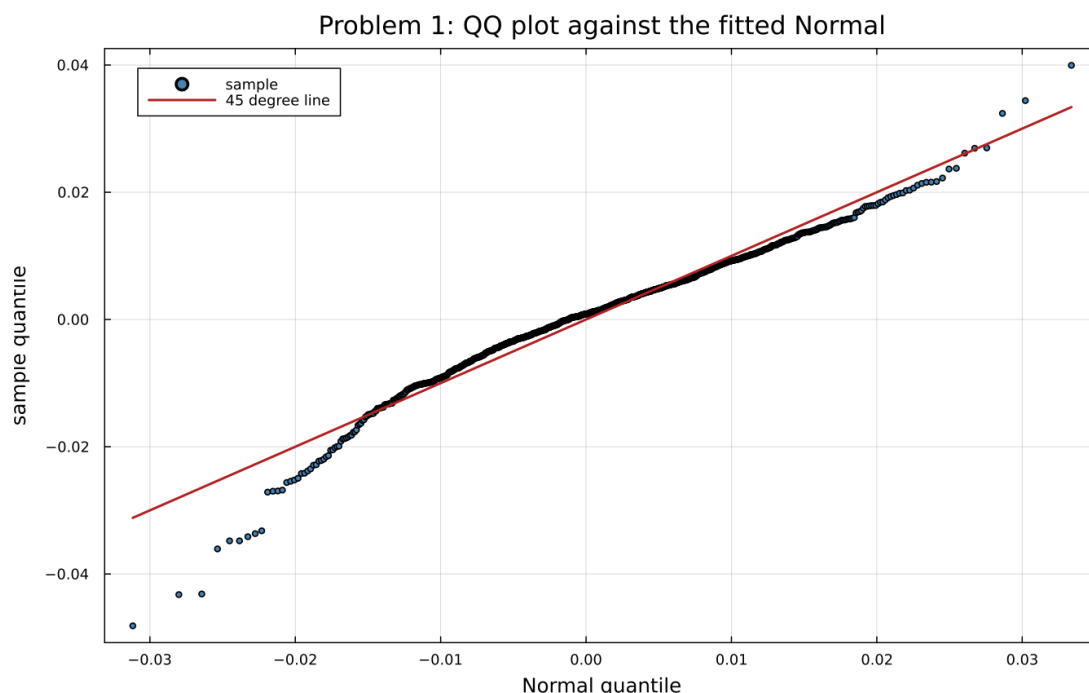


Figure 2: The same comparison as a QQ plot

1.3 Reconcile

Part (c). The fitted Normal is wrong in three places, in three different directions. Counting observations past each of its quantiles:

Quantile	Left observed	Expected	Right observed	Expected
0.5%	18	5.0	5	5.0
1%	26	10.0	6	10.0
2.5%	40	25.0	15	25.0
5%	50	50.0	34	50.0
10%	83	100.0	78	100.0

The left tail is far too thin, and it gets worse the further out you go: 1.6 times too few at 2.5%, 2.6 times at 1%, 3.6 times at 0.5%. This is the error that matters. The empirical 1% quantile is -0.02699 against the model's -0.02172 , so a 1% VaR read off this Normal is **24% too small**. The worst observation is -0.0482 , which is 5.02 fitted standard deviations out.

The right tail is too thick in the model, not too thin. At 1% the Normal expects 10 large gains and there are 6. A symmetric fit splits the difference between a heavy left tail and a light right tail, so it overstates the upside by as much as it understates the downside.

The centre is too fat in the data. 732 observations sit within one fitted standard deviation against 683 expected. Matching the variance forces $\hat{\sigma}$ up to accommodate the few large moves, and that inflated $\hat{\sigma}$ is then too wide for the quiet days.

So the fitted Normal is too wide in the body, too narrow in the left tail, and too wide in the right tail.

2 A Regression Whose Errors Are Not Normal

2.1 Predict

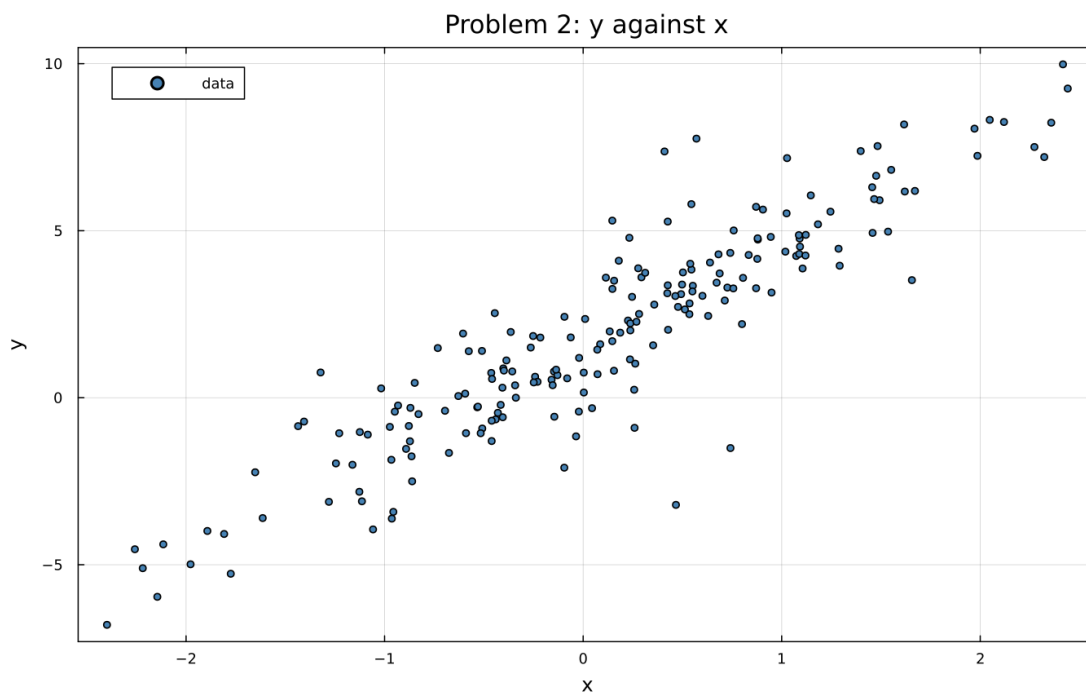


Figure 3: The scatter, before fitting anything

Part (a). The relationship is linear and strong, with no curvature and no fanning out: the vertical spread looks the same at $x = -2$, $x = 0$ and $x = +2$, so the error variance looks constant. The slope is around 3 and the line does not pass through the origin.

What stands out is a handful of points well off the line, near $(0.47, -3.2)$, $(0.75, -1.5)$, $(0.41, +7.4)$ and $(0.57, +7.8)$. They are in the middle of the x range, not at the edges, so they are not leverage points, and they miss by three or four times the typical spread. A tight linear body with a few large vertical misses on both sides is what a **fat tailed, roughly symmetric** error looks like. I expect a **Student's t** error with ν somewhere between 3 and 6.

Part (b). The violated assumption is **assumption 7**, normally distributed errors. The others hold: the cloud is straight (1), an intercept is included (2), the misses are not concentrated at particular x (3), these are cross sectional draws (4), the spread does not widen with x (5), and there is one regressor plus a constant (6).

Effect on the slope: almost none. Unbiasedness and consistency of $\hat{\beta} = (X'X)^{-1}X'Y$ come from assumptions 1 through 6, all of which hold. What assumption 7 buys is that OLS is also the maximum likelihood estimator, and that is what we lose. I expect the slope to move by well under half a standard error when refitted under a t likelihood.

Effect on the standard error: this is the damage. Least squares weights each observation by the square of its residual, so a miss of 6 contributes 36 units of evidence about the slope. Under a normal error that is correct. Under a fat tailed error it is not, because a miss of 6 is an ordinary event carrying little information. OLS therefore lets a few tail draws push the line around and is

genuinely noisier than an estimator that discounts them, which a t likelihood does automatically. I expect $\text{se}(\hat{\beta})$ under OLS to be visibly larger than under the t MLE, somewhere in the 10% to 30% range.

I am not predicting that the OLS standard error is biased downward. That is what happens when assumption 4 breaks. Here the interval is honest about the estimator we chose; the estimator is just worse than it needs to be.

2.2 Fit

$n = 200$.

OLS.

	Estimate	Std. error	t
α	1.560740	0.096269	16.21
β	2.990757	0.096640	30.95
s	1.351401		

$R^2 = 0.8287$. Durbin Watson = 2.0159, so assumption 4 is fine. Smallest eigenvalue of $X'X$ is 175.05, so there is no multicollinearity. Residual skewness -0.362 and **excess kurtosis** $+3.617$.

MLE, Normal error. $\alpha = 1.560740$, $\beta = 2.990757$, $\sigma = 1.344627$, log likelihood -343.0110 . The coefficients match OLS to every printed digit, which is the Week 2 result that the two problems coincide under a normal error. Only σ differs, because MLE divides by n and OLS by $n - p$.

MLE, Student's t error.

	Estimate	Std. error
α	1.547979	0.080660
β	3.019179	0.078675
σ (scale)	0.935509	
ν	3.6322	
Log likelihood	-328.5612	

AICc, with $k = p + d$: three parameters for the Normal models, four for the t.

Model	Log likelihood	k	AICc	BIC
OLS	-343.0110	3	692.1445	701.9170
MLE, Normal	-343.0110	3	692.1445	701.9170
MLE, t	-328.5612	4	665.3275	678.3156

OLS and the Normal MLE are the same model fitted two ways, so they share a row. **AICc selects the t by 26.82.**



Figure 4: Both fitted lines over the data

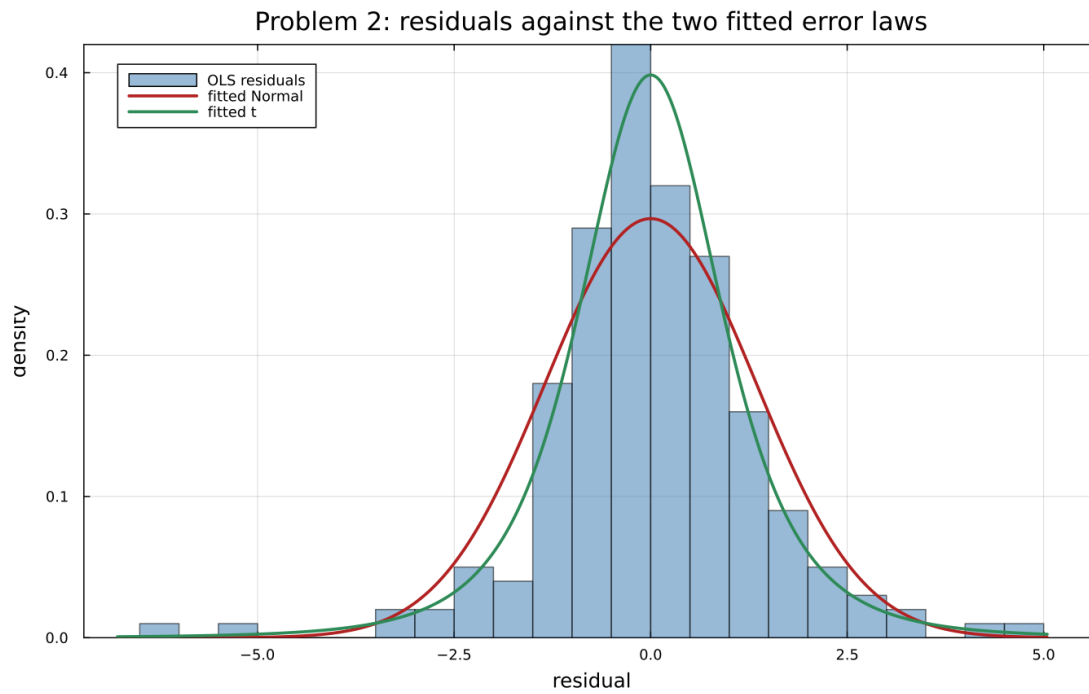


Figure 5: Residuals against the two fitted error laws

2.3 Reconcile

Part (c). OLS 2.990757, Normal MLE 2.990757, t MLE 3.019179. The spread is 0.0284, which is 0.95% of the slope and 0.294 **OLS standard errors**, so the violation moved the slope about as little as predicted.

The standard errors are 0.096640 for OLS against 0.078675 for the t, a ratio of 1.228: the OLS interval is **23% wider**, inside the 10% to 30% range I predicted. The t likelihood recognises that a residual of -4 is an ordinary draw from a $\nu = 3.6$ error and refuses to let it drag the line, which buys a sharper estimate from the same 200 observations without collecting more data.

Part (d). It is getting the **error distribution** wrong, and with it every probability statement built on that distribution. The Normal has one parameter doing two jobs, setting the width of the body and the weight of the tail. To accommodate four residuals past ± 4 it must inflate σ to 1.345, and that single number is then applied everywhere. The t splits the jobs: scale 0.936 for the body, $\nu = 3.63$ for the tail.

Counting how often the residuals fall outside each model's own central interval:

Interval	Expected outside	Outside the Normal	Outside the t
90%	20.0	17	21
95%	10.0	9	9
99%	2.0	4	2
99.5%	1.0	4	1

The Normal is fine at 90% and 95% and then breaks exactly where a risk manager starts paying attention.

Part (e).

Level	Normal	t	Wider
95%	2.2117	2.0538	Normal
99%	3.1281	3.7098	t
99.5%	3.4635	4.6201	t
99.9%	4.1552	7.4473	t

The two cross at about the **97.0%** quantile. The reason is the parameter counting above. The Normal's scale is 1.345 against the t's 0.936, 44% larger, and at the 95% level the multipliers are 1.96 against 2.20, so the scale advantage more than covers the gap and the Normal is wider. At 99.5% the multipliers are 2.576 against 4.939, the t's has nearly doubled while the Normal's grew by a third, and the scale advantage is overwhelmed.

For a capital buffer I would want the t. A buffer is set at 99% or beyond, which is precisely where the Normal is the narrower and therefore the more dangerous of the two. Sizing off the Normal gives 3.46 where the correctly specified model says 4.62, a **25% shortfall**, and the count above shows the Normal's 99.5% level is breached four times where it should be breached once. The Normal is conservative in the region where being conservative costs nothing and aggressive in the region where being aggressive means being undercapitalised, which is the wrong way round.

3 Pearson Against Spearman

3.1 Predict

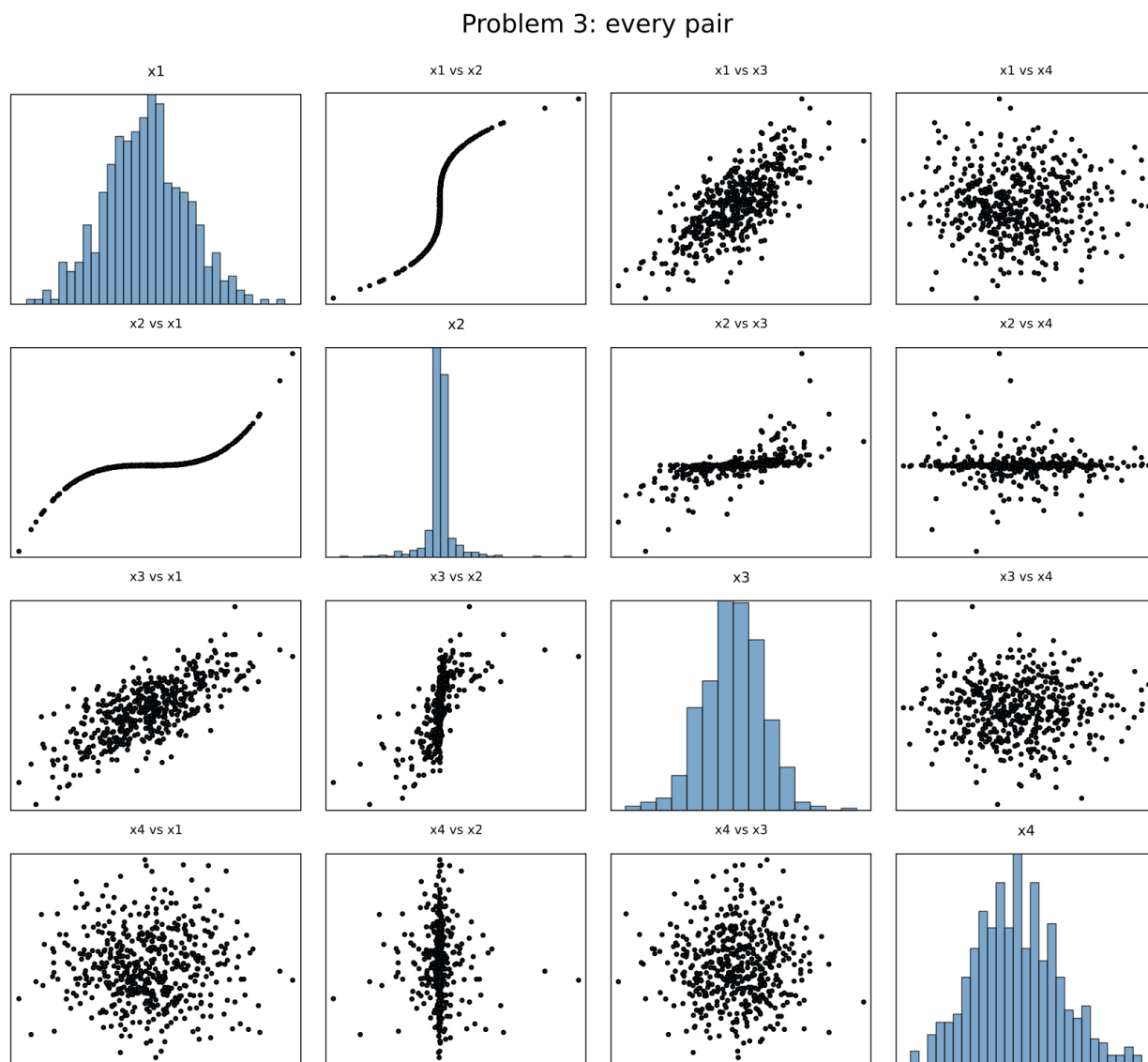


Figure 6: Every pair

Part (a).

Biggest disagreement: x_1 and x_2 . The scatter is a clean, strictly increasing S curve with almost no noise around it, which is a cubic. Spearman only looks at order and the relationship is monotone, so it should be close to +1. Pearson measures closeness to a **straight line**, and a cubic is nothing like a line at its ends, so it will be pulled well below. I expect Spearman above 0.95 and Pearson around 0.8.

Second, smaller: x_2 and x_3 . x_3 looks linearly related to x_1 and x_2 is a monotone nonlinear function of x_1 , so the same mechanism operates on this pair, diluted by the noise in the x_1 to x_3

link.

Agree: everything involving x_4 , whose panels are featureless blobs, so both coefficients sit near zero where they cannot disagree by much. **And x_1 with x_3** , an ordinary elliptical cloud with a linear tilt, which is the case the two measures were built to agree on. There I expect Spearman slightly *below* Pearson, since ranking a genuinely linear relationship throws away a little information.

3.2 Fit

Part (b). $n = 500$. Pearson:

	x_1	x_2	x_3	x_4
x_1	1.0000	0.7991	0.7195	-0.0047
x_2	0.7991	1.0000	0.5876	-0.0209
x_3	0.7195	0.5876	1.0000	-0.0086
x_4	-0.0047	-0.0209	-0.0086	1.0000

Spearman:

	x_1	x_2	x_3	x_4
x_1	1.0000	0.9719	0.6839	0.0002
x_2	0.9719	1.0000	0.6674	-0.0011
x_3	0.6839	0.6674	1.0000	-0.0024
x_4	0.0002	-0.0011	-0.0024	1.0000

Gaps, Spearman minus Pearson:

Pair	Pearson	Spearman	Gap
x_1, x_2	0.7991	0.9719	+0.1728
x_2, x_3	0.5876	0.6674	+0.0798
x_1, x_3	0.7195	0.6839	-0.0356
x_2, x_4	-0.0209	-0.0011	+0.0198
x_3, x_4	-0.0086	-0.0024	+0.0062
x_1, x_4	-0.0047	0.0002	+0.0049

The pair with the largest gap is x_1, x_2 at +0.1728. The ordering held, including the sign flip on x_1, x_3 , which is the only negative gap.

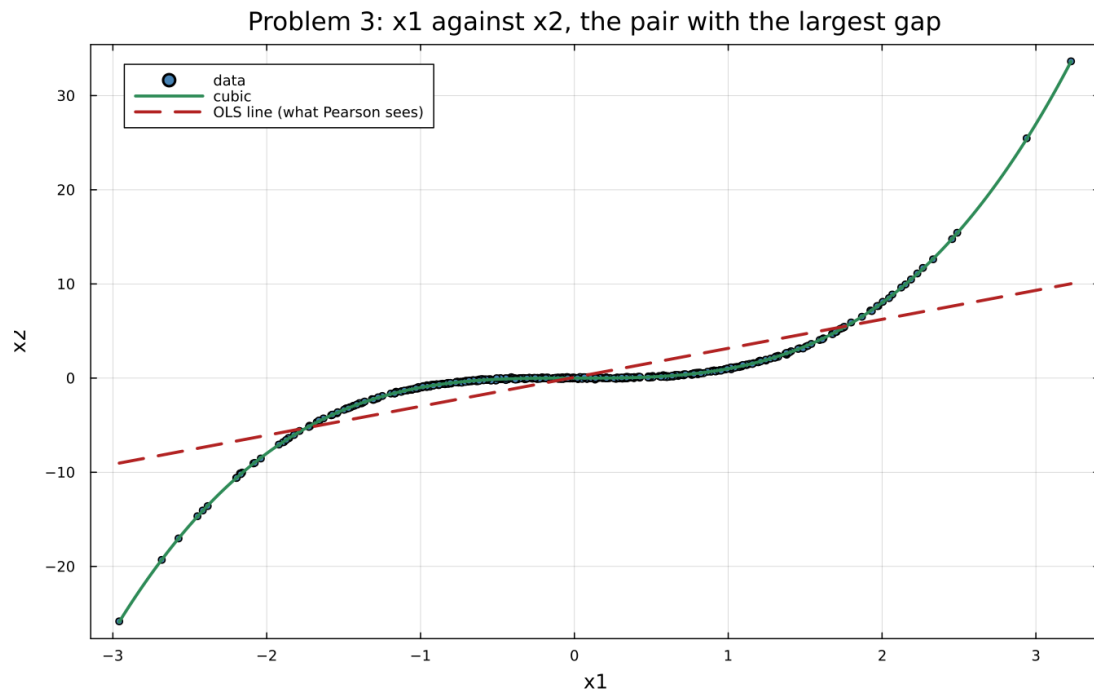


Figure 7: The pair with the largest gap

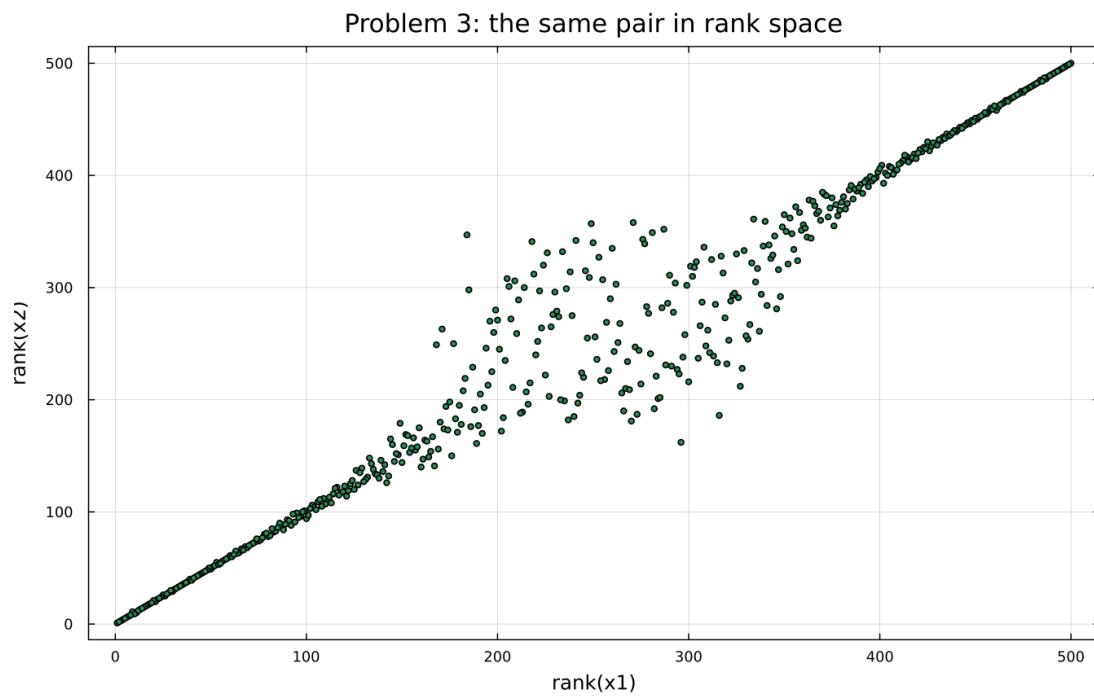


Figure 8: The same pair in rank space

3.3 Reconcile

Part (c). Confirming the functional form first: $\text{cor}(x_1^3, x_2) = 0.999917$, and regressing x_2 on x_1^3 gives $x_2 = -0.0023 + 0.9994 x_1^3$ with $R^2 = 0.99983$ and a residual standard deviation of 0.0502 against 3.894 for x_2 itself. So x_2 is a cube of x_1 plus a whisper of noise.

Pearson is a statement about a straight line. The best line through a cubic is the dashed line in the figure: above the points at the left, below them at the right, and cutting the flat middle at the wrong slope. $\rho^2 = 0.638$, so a linear rule captures 64% of the variance of x_2 and misses 36%. *Pearson* is not making an error; it is correctly reporting that a linear predictor does a mediocre job.

Spearman replaces both series by ranks, and a strictly increasing transform leaves ranks untouched, so the curvature disappears and the rank scatter is a near perfect diagonal.

It is 0.9719 rather than exactly 1 because of the noise, and specifically because a cubic has zero derivative at the origin. Near $x_1 = 0$ the values of x_2 are packed into a tiny interval and a residual of 0.05 reorders dozens of points; out where the cubic is steep the same noise cannot reorder anything. The mean absolute rank difference is 43.2 for $|x_1| < 0.5$, 6.0 for $0.5 \leq |x_1| < 1$, and 0.6 beyond that. All of the lost 0.028 comes from the flat middle.

Spearman is the honest description of this pair. The relationship is very nearly deterministic, and reporting it as 0.80 understates the dependence badly. A Pearson of 0.80 suggests there is real diversification available between these two series, and there is essentially none.

Pearson is answering a narrower question: how much of the variance of x_2 a linear function of x_1 can explain. The answer is 64%, and 0.7991 is the right number for that. It is what you want if you are hedging x_2 with a fixed quantity of x_1 , because the residual risk really would be 36% of the variance, and it is the number that belongs in a covariance matrix, since $w'\Sigma w$ is a statement about linear combinations only. The mistake is not picking the wrong coefficient but quoting one and believing you were told the other.

4 Conditional Distributions

4.1 Predict

$n = 1000$, and the sample covariance matrix only:

$$\hat{\mu} = \begin{bmatrix} \mu_{x_1} \\ \mu_{x_2} \end{bmatrix} = \begin{bmatrix} -0.016714 \\ +0.045288 \end{bmatrix}, \quad \begin{bmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{bmatrix}, \quad \rho = 0.798623.$$

Part (a). The Week 2 result conditions block 1 on block 2, so I order the vector with the series I want first and the series I observe second:

$$X = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} 4.104749 & 1.696902 \\ 1.696902 & 1.099875 \end{bmatrix}.$$

So $\Sigma_{11} = \text{var}(x_2)$ is the block for the conditioned series, $\Sigma_{22} = \text{var}(x_1)$ the block for the observed one, and $\Sigma_{12} = \Sigma_{21} = \text{cov}(x_1, x_2)$ the cross blocks.

$$\bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = 4.104749 - \frac{1.696902^2}{1.099875} = \mathbf{1.486746},$$

conditional standard deviation 1.219322. The factor by which uncertainty falls is

$$\frac{\bar{\Sigma}}{\Sigma_{11}} = 1 - \frac{\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}}{\Sigma_{11}} = 1 - \rho^2,$$

which here is $1.486746/4.104749 = \mathbf{0.362201}$, matching $1 - 0.798623^2$ to six decimals. Learning x_1 removes **63.78% of the variance**. In standard deviation terms the fall is smaller, $\sqrt{0.3622} = 0.6018$, a **39.82% reduction**, and it is the standard deviation that sets the width of an interval.

Part (b). No. The term that settles it is the absence of a from $\bar{\Sigma}$. Writing the two halves side by side,

$$\bar{\mu} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2), \quad \bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21},$$

the conditioning value appears once, inside $(a - \mu_2)$ in $\bar{\mu}$, and **nowhere in $\bar{\Sigma}$** , which is built entirely from constant blocks of Σ . Under the multivariate normal, observing an extreme x_1 shifts the centre of the distribution of x_2 and does not widen or narrow it at all. That is why the band below is a pair of parallel lines rather than a curve.

I doubt the data will cooperate. Given that Problems 1 and 2 were both fat tailed, I expect coverage below 95% and a band that is too narrow where x_1 is far from its mean.

4.2 Fit

Part (c).

$$\bar{\mu} = 0.045288 + \frac{1.696902}{1.099875}(a + 0.016714) = 0.045288 + 1.542813(a + 0.016714).$$

The coefficient on x_1 is $\Sigma_{12}\Sigma_{22}^{-1} = 1.542813$, which in regression terms is the **OLS slope of x_2 on x_1** . The single regressor form $\hat{\beta} = \text{cov}(x, y)/\sigma_x^2$ is that expression written without matrix notation, and running the regression confirms it:

	From Σ	From OLS
Slope	1.542813	1.542813 (se 0.036803)
Intercept	0.071073	0.071073
Residual variance	$\bar{\Sigma} = 1.486746$	$s^2 = 1.488235$

The residual variances differ only by the n against $n - p$ convention. The 95% band is $\bar{\mu} \pm 1.959964\sqrt{\bar{\Sigma}} = \bar{\mu} \pm 2.389827$, the same half width everywhere.

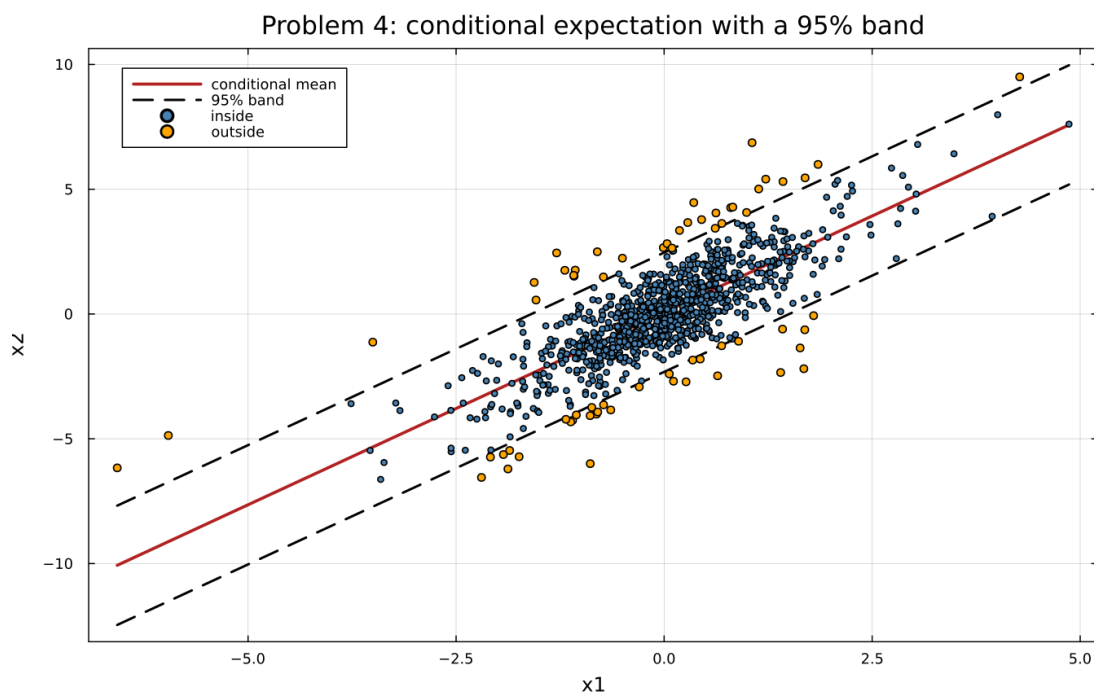


Figure 9: Conditional expectation with a 95% band

Part (d). 935 of 1000 inside, coverage 0.9350 against a nominal 0.9500. The binomial standard error is 0.0078, so the shortfall is a little under two standard errors.

Part (e).

Bucket	n	Coverage	Std. err.	Resid. sd
$ x_1 - \mu < 1 \text{ sd}$	759	0.9565	0.0074	1.0758
$1 \leq x_1 - \mu < 2 \text{ sd}$	190	0.8579	0.0253	1.5804
$ x_1 - \mu \geq 2 \text{ sd}$	51	0.9020	0.0416	1.5957
All	1000	0.9350	0.0078	1.2193

4.3 Reconcile

Part (f). Coverage is not flat, and the residual standard deviation column says why. Inside one standard deviation the residual scale is 1.076; beyond it, 1.58 to 1.60. The residuals get **roughly 48% wider** as x_1 moves away from its mean, so a fixed band calibrated to the pooled scale of 1.219 is too wide in the crowded middle, where it over covers at 95.65%, and too narrow in the wings, where it under covers at 85.79%. Regressing $|\text{residual}|$ on $|x_1 - \mu|$ gives a slope of +0.2660 with a standard error of 0.0342, so $t = +7.78$. That is not noise.

Coverage rises again to 90.20% in the outermost bucket, which looks like the pattern reversing. It is not. That bucket has 51 observations and a standard error of 4.2 points, so 90.2% and 85.8% are about one standard error apart. The residual scale, which is estimated more stably, rises monotonically across all three. The middle bucket, with 190 observations and a 2.5 point standard error, is 5.6 standard errors below nominal on its own, and that single bucket is the whole result.

The assumption that failed is joint normality, and specifically the property from part (b) that $\bar{\Sigma}$ contains no a . The band has constant width only because the multivariate normal promises a conditional variance that does not depend on the conditioning value, and the data breaks that promise: the conditional variance here rises with $|x_1 - \mu|$. In the OLS vocabulary this is **assumption 5**, constant error variance. In the multivariate normal vocabulary it means no bivariate normal, at any μ and Σ , could have generated this sample.

There is a second, smaller failure. The residuals have excess kurtosis +1.51 pooled and 0.98, 1.06, 0.33 within the three buckets, so even after letting the width vary the conditional law is fatter tailed than a normal and 1.96 is the wrong multiplier.

What survives from part (c).

1. *The line.* $\Sigma_{12}\Sigma_{22}^{-1} = 1.542813$ is the least squares projection of x_2 on x_1 , which is defined by second moments alone and needs no normality and no homoskedasticity. It is still the best linear predictor and still unbiased and consistent, because OLS assumptions 1 through 4 and 6 hold. What it loses is the guarantee that it equals the true conditional expectation $E[x_2 | x_1]$; that equality came from joint normality.
2. *The value of $\bar{\Sigma}$ as an average.* 1.486746 is still the correct unconditional variance of the projection residual, the average of the true conditional variances over the distribution of x_1 . That is why overall coverage lands at 93.5%, close to nominal.
3. *The decomposition $\bar{\Sigma}/\Sigma_{11} = 1 - \rho^2 = 0.3622$,* another second moment identity.

What does not.

1. *The constant width of the band.* It should be a widening cone.
2. *The claim that $\bar{\Sigma}$ is the conditional variance at any particular x_1 .* It is the average, not the value; at $x_1 = 3$ it understates the spread by about a third.
3. *The 1.96 multiplier,* which presumes a normal conditional law.
4. *Reading $\bar{\mu}$ as the conditional expectation* rather than as the linear projection.

Half widths each bucket would have needed for 95% coverage:

Bucket	Needed	Supplied	Ratio
$ x_1 - \mu < 1 \text{ sd}$	2.2839	2.3898	0.96
$1 \leq x_1 - \mu < 2 \text{ sd}$	3.3253	2.3898	1.39
$ x_1 - \mu \geq 2 \text{ sd}$	3.5684	2.3898	1.49

Bucket	Needed	Supplied	Ratio
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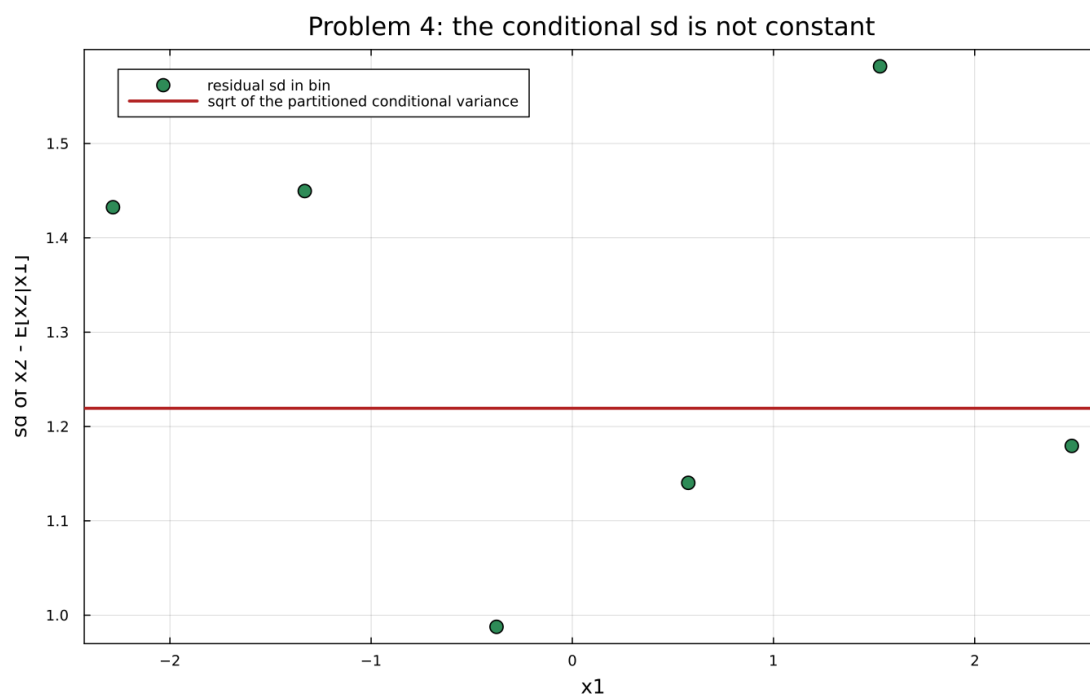


Figure 10: Residual standard deviation by bin of x_1

Coverage came in below 95% and the band was too narrow in the wings, as predicted. What I did not anticipate was that the shortfall would sit almost entirely in the middle bucket, and that the outermost bucket would be too small to say anything on its own.

5 Identifying an AR or MA Order

5.1 Predict

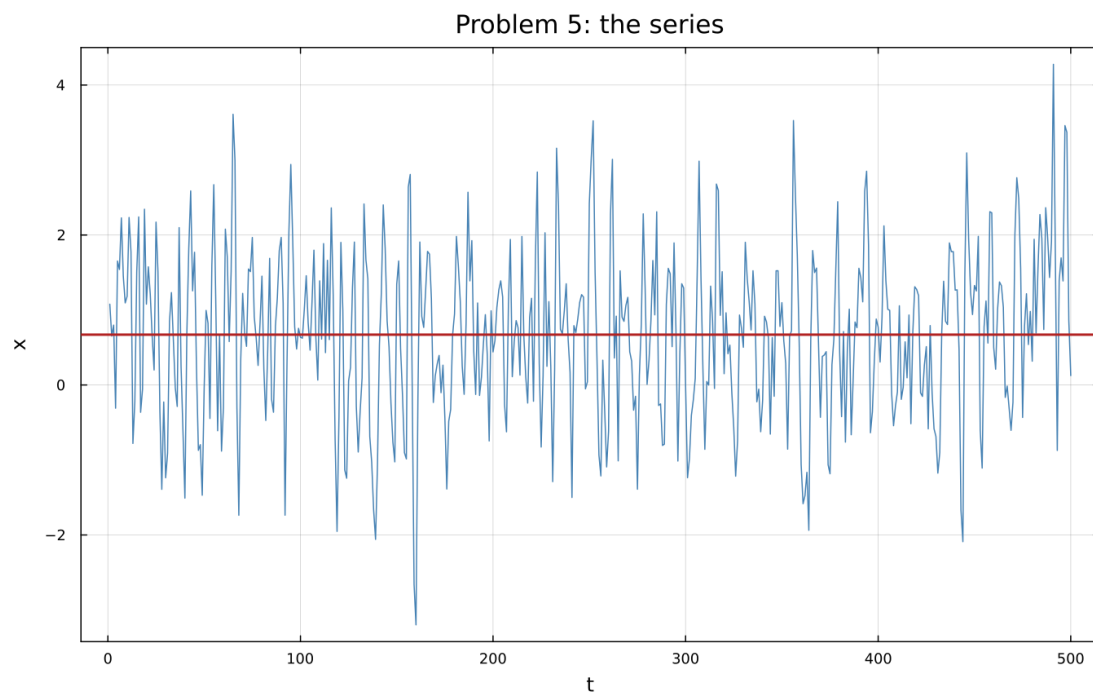


Figure 11: The series

Problem 5: ACF decays, PACF cuts off

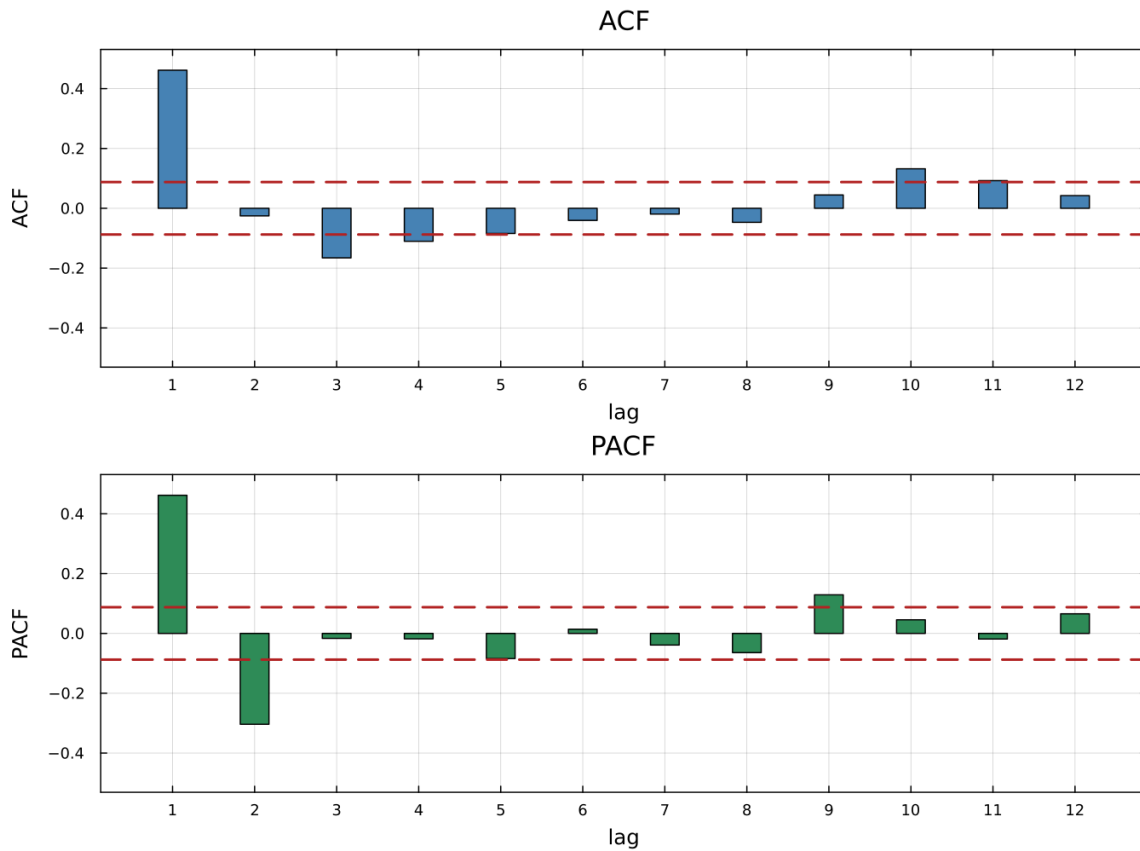


Figure 12: ACF and PACF with the significance band

The series is stationary: it oscillates around a constant level of about 0.67 with no drift in amplitude and no trend.

Lag	ACF	PACF
1	+0.4616	+0.4619
2	−0.0254	−0.3037
3	−0.1657	−0.0166
4	−0.1104	−0.0182
5	−0.0841	−0.0841
6	−0.0405	+0.0140
7	−0.0193	−0.0389
8	−0.0469	−0.0642
9	+0.0446	+0.1291
10	+0.1321	+0.0455

Part (a). The PACF cuts off, the ACF decays, which by the Week 2 identification table is an **autoregressive** process, and the PACF cuts off after **lag 2**. I commit to **AR(2)** before fitting.

The PACF has two large values, $+0.462$ and -0.304 , then falls off a cliff: lags 3 through 8 come in at -0.017 , -0.018 , -0.084 , $+0.014$, -0.039 , -0.064 , all inside the band. Dropping from -0.304 to -0.017 in one step is a cut off, not a decay.

The ACF does the opposite. It starts at $+0.462$, passes through roughly zero at lag 2, then runs negative for six consecutive lags, shrinking as it goes. That damped oscillation is what an AR(2) with a positive first and negative second coefficient produces. An MA(q) would have to be exactly zero beyond lag q , and this is not zero after any lag.

The Yule Walker equations give the coefficients from the first two autocorrelations: $\phi_2 = (\rho_2 - \rho_1^2)/(1 - \rho_1^2) = -0.303$ and $\phi_1 = \rho_1(1 - \phi_2) = 0.601$, so I expect roughly $(0.60, -0.30)$.

Part (b). The band is $\pm 1.96/\sqrt{n} = \pm 1.96/\sqrt{500} = \pm 0.087654$, the region marked on the Week 2 example figures. It comes from Bartlett's result that under the null of white noise a sample autocorrelation is approximately $N(0, 1/n)$, so a value outside it is significant at 5%.

The rule I used: the PACF cuts off at lag p if it is outside the band at every lag up to p and inside the band for a run of lags immediately after. Lags 1 and 2 are 5.3 and 3.5 times the band; lags 3 through 8 are all inside it.

Two caveats. The band tests one lag at a time, so across 12 lags about 0.6 false positives are expected. There is exactly one candidate, the PACF at lag 9 of $+0.1291$, and I treat it as that rather than as evidence for an AR(9), since nothing would produce a ninth order dependence with nothing at lags 3 through 8. Second, the band is not the whole rule: the ACF is outside it at lags 3 and 4 too, and applying "cuts off where it stops being significant" mechanically would call this an MA(4). That reading fails because the ACF is decaying through the band rather than cutting off, and the shape of the sequence matters as much as which side of the line each value falls on.

5.2 Fit

Part (c). All six are fitted by **exact Gaussian maximum likelihood** using the Week 2 form

$$ll = -\frac{1}{2} (n \ln 2\pi + \ln|\Gamma| + X'\Gamma^{-1}X), \quad X = Y - \mu,$$

with Γ built from the model implied autocovariances: Yule Walker plus upward recursion for the AR models, $\gamma(k) = \sigma^2 \sum_i \theta_i \theta_{i+k}$ for the MA models. I used the exact likelihood rather than a conditional one because a conditional likelihood drops the first p observations, which would score AR(1) on 499 observations and AR(3) on 497, and information criteria computed on different samples are not comparable. This way all six models use all 500.

$k = p + d$: the number of coefficients, plus one for the mean, plus one for σ^2 .

Model	Parameters	k	Log likelihood	AICc	BIC
AR(1)	$\phi = (0.4611)$	3	-706.2693	1418.5870	1431.1824
AR(2)	$\phi =$ $(0.6012, -0.3026)$	4	-682.1788	1372.4383	1389.2159
AR(3)	$\phi =$ $(0.5962, -0.2927, -0.0165)$	5	-682.1109	1374.3432	1395.2948
MA(1)	$\theta = (0.5296)$	3	-691.5179	1389.0843	1401.6797

Model	Parameters	k	Log likelihood	AICc	BIC
MA(2)	$\theta =$ (0.6314, 0.1615)	4	−686.5754	1381.2316	1398.0092
MA(3)	$\theta =$ (0.5931, 0.0648, −0.1555)	5	−681.9975	1374.1165	1395.0681

Since the likelihood is written from the formula rather than taken from a package, the AR(2) coefficients are cross checked two ways:

Method	ϕ_1	ϕ_2
Yule Walker from the sample ACF	+0.60156	−0.30312
Conditional least squares	+0.60236	−0.30375
Exact Gaussian MLE	+0.60120	−0.30264

5.3 Reconcile

Part (d). **AICc selects AR(2)** at 1372.4383. MA(3) is next at 1374.1165, a gap of 1.68, and AR(3) third at 1374.3432. BIC agrees and, with its larger per parameter penalty, more emphatically, putting AR(2) ahead of MA(3) by 5.85. **This matches the prediction**, and the Yule Walker coefficient guess of (0.60, −0.30) matched the fitted (0.6012, −0.3026).

MA(3) nearly wins, which is the identification table doing its job: an AR(2) is an infinite order MA, so a finite MA needs several terms to approximate it and by three terms has nearly succeeded. It gets there by spending an extra parameter for a fit AR(2) already had. I would still take AR(2), because it is the smaller model, because BIC prefers it by 5.85, and because the diagnostic pointed at AR before either was fitted.

Part (e).

	AR(2)	AR(3)
ϕ_1	0.60120	0.59621
ϕ_2	−0.30264	−0.29269
ϕ_3		− 0.01655
Log likelihood	−682.1788	−682.1109
AICc	1372.4383	1374.3432

The third coefficient is −0.01655, about one fiftieth the size of the other two, and it buys a log likelihood improvement of 0.0679. An extra parameter costs about 2.0 in AICc terms and needs a gain above roughly 1.0 to pay for itself, so this one delivers about a fifteenth of the entry price and AICc gets worse by 1.90. The likelihood ratio statistic is $2(\ell_3 - \ell_2) = 0.1357$ against a $\chi^2(1)$, a p value of 0.71. The trade AICc is making is to accept a parameter only if its improvement exceeds what an irrelevant parameter would deliver by chance, which is about 0.5 in expectation. This one delivered less than that.

R^2 **would not have made the same choice**, because it has no price list:

Model	In sample R^2	Residual sd
AR(1)	0.213262	0.995244
AR(2)	0.285872	0.949159
AR(3)	0.286158	0.949912

R^2 **risers** from AR(2) to AR(3). It has to: AR(2) is AR(3) with ϕ_3 constrained to zero, so the AR(3) fit maximizes over a strictly larger set and cannot come out worse. Adding a column of random numbers would raise it too. So R^2 ranks AR(3) first and would keep ranking bigger models first however many useless lags were added. The increase is 0.000286, which is the honest measure of what the third lag contributed and is indistinguishable from zero. AICc, adjusted R^2 and BIC are all R^2 with an invoice attached, and the invoice is what stops you selecting a model that describes this sample beautifully and forecasts nothing.